

Optimal Transport and the Wasserstein Distance

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Setting

Let $(\mathcal{X}, d)$ be a Polish metric space (e.g. $\mathbb{R}^n$ with the Euclidean metric) and let $\mathcal{P}_p(\mathcal{X})$ denote the set of Borel probability measures on $\mathcal{X}$ with finite p -th moment, for some $p \geq 1$.

Definition 1 (Coupling / transport plan). Given $\mu, \nu \in \mathcal{P}_p(\mathcal{X})$, a *coupling* of μ and ν is a Borel probability measure π on $\mathcal{X} \times \mathcal{X}$ whose marginals are μ and ν :

$$\pi(A \times \mathcal{X}) = \mu(A), \quad \pi(\mathcal{X} \times B) = \nu(B).$$

We write $\pi \in \Pi(\mu, \nu)$. The set $\Pi(\mu, \nu)$ is always non-empty (contains $\mu \otimes \nu$) and weak-* compact.

Definition 2 (Wasserstein- p distance, Kantorovich formulation).

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\pi(x, y) \right)^{1/p}.$$

By weak-* compactness of $\Pi(\mu, \nu)$ and lower semi-continuity of the cost, the infimum is attained.

Definition 3 (Monge formulation). A *transport map* $T : \mathcal{X} \rightarrow \mathcal{X}$ pushes μ forward to ν if $T_{\#}\mu = \nu$, i.e. $\nu(B) = \mu(T^{-1}(B))$ for every Borel B . The Monge problem is

$$\inf_{T_{\#}\mu = \nu} \int_{\mathcal{X}} d(x, T(x))^p d\mu(x).$$

Every map T induces a (deterministic) coupling $(\text{id}, T)_{\#}\mu$, so the Monge value is always $\geq$ the Kantorovich value. Under regularity (e.g. μ absolutely continuous on $\mathbb{R}^n$, $p > 1$), Brenier's theorem gives equality and a unique optimal map.

The 1D case: a closed-form theorem

In one dimension everything collapses to ordering quantiles. Recall that for a probability measure μ on $\mathbb{R}$ with c.d.f. $F_{\mu}(x) = \mu((-\infty, x])$, the *quantile function* is

$$F_{\mu}^{-1}(u) = \inf\{x \in \mathbb{R} : F_{\mu}(x) \geq u\}, \quad u \in (0, 1).$$

If $U \sim \text{Unif}(0, 1)$ then $F_{\mu}^{-1}(U) \sim \mu$.

Theorem 4 (1D Wasserstein closed form). *Let $\mu, \nu \in \mathcal{P}_p(\mathbb{R})$ with $p \geq 1$. Then*

$$W_p(\mu, \nu)^p = \int_0^1 |F_{\mu}^{-1}(u) - F_{\nu}^{-1}(u)|^p du.$$

Moreover the infimum is attained by the monotone coupling $\pi^ = (F_{\mu}^{-1}, F_{\nu}^{-1})_{\#}\text{Leb}_{(0,1)}$.*

Proof sketch. Upper bound. The map $u \mapsto (F_\mu^{-1}(u), F_\nu^{-1}(u))$ from $(0, 1)$ to $\mathbb{R}^2$ pushes Lebesgue measure on $(0, 1)$ to a probability π^* on $\mathbb{R}^2$. Its first marginal is the law of $F_\mu^{-1}(U) = \mu$, and similarly for ν , so $\pi^* \in \Pi(\mu, \nu)$. Hence

$$W_p(\mu, \nu)^p \leq \int_{\mathbb{R}^2} |x - y|^p d\pi^*(x, y) = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)|^p du.$$

Lower bound (the key step). The cost $c(x, y) = |x - y|^p$ for $p \geq 1$ satisfies the *Monge condition* (a.k.a. submodularity): for all $x < x'$ and $y < y'$,

$$c(x, y) + c(x', y') \leq c(x, y') + c(x', y).$$

(For $p = 1$ this is immediate: both sides equal $|x - y'| + |x' - y|$ in the “crossing” configuration but $(x' - x) + (y' - y)$ vanishes when one rearranges; the general case follows from convexity of $t \mapsto |t|^p$.)

Now let π be any coupling and consider any pair of points $(x, y), (x', y') \in \text{supp}(\pi)$ with $x < x'$ and $y > y'$. The Monge condition says we can “uncross” this pair: replace mass $\varepsilon > 0$ on these two pairs with mass ε on (x, y') and (x', y) instead, without changing either marginal, and the total cost can only decrease. An exchange/measurable-selection argument (a swapping/cyclical-monotonicity lemma; see Villani Ch. 5) shows that any coupling can be made *monotone* (i.e. supported on a non-decreasing set) without increasing cost. But the only monotone coupling with the prescribed marginals is π^* . Hence π^* is optimal, proving the lower bound. $\square$

Remark. The theorem yields the trivial empirical-distribution algorithm: to estimate W_p between samples $x_1, \dots, x_n$ and $y_1, \dots, y_n$, sort each list and compute $(\frac{1}{n} \sum_i |x_{(i)} - y_{(i)}|^p)^{1/p}$.

Examples

Two point masses. $W_p(\delta_a, \delta_b) = |a - b|$ for every p .

Translated Gaussians. $\mu = \mathcal{N}(0, 1), \nu = \mathcal{N}(m, 1)$: $W_p(\mu, \nu) = |m|$. Compare with $D_{\text{KL}}(\mu \parallel \nu) = m^2/2$.

Pointers

The Kantorovich–Rubinstein duality $W_1(\mu, \nu) = \sup_{\|f\|_{\text{Lip}} \leq 1} \int f d(\mu - \nu)$ is the gateway to Wasserstein GANs. Brenier’s theorem identifies the optimal transport map for $p = 2$ on $\mathbb{R}^n$ as the gradient of a convex function. See Villani’s *Optimal Transport: Old and New* and Peyré–Cuturi’s *Computational Optimal Transport*.